

Multiple Choice (10 marks)

1. There are 230 calories in 4 ounces of a type of ice cream. How many calories are in 6 ounces of that ice cream?
 - (a) 232
 - (b) 236
 - (c) 345
 - (d) 460
2. A zoo has 15 toucans and 60 parrots. What is the ratio of the number of toucans to the number of parrots at the zoo?
 - (a) 1:04
 - (b) 1:05
 - (c) 4:01
 - (d) 4:05
3. Balloons come in packages of 8. If you need 42 balloons, how many packages should you buy?
 - (a) 5
 - (b) 6
 - (c) 8
 - (d) 42
4. Suppose two events, E and F, have nonzero probabilities p and q , respectively. Which of the following is impossible?
 - (a) $p + q > 1$
 - (b) $p - q < 0$
 - (c) $p / q > 1$
 - (d) E and F are both independent and mutually exclusive.
5. Let p , q , and r be constants. One solution to the equation $(x - p)(x - q) = (r - p)(r - q)$ is $x = r$. Find the other solution in terms of p , q , and r .
 - (a) pqr
 - (b) $pq-r$
 - (c) $p-q-r$
 - (d) $p+q-r$

True / False (10 marks)

Answer True or False

6. True or False: To find the number of students in each group, you should add the total number of students and the number of groups, represented by $30+5$.
7. True or False: The smallest positive integer n greater than 1, whose positive divisors multiply to n^6 , is 30.
8. True or False: The greatest common factor of 252 and 96 is 24.
9. True or False: Randomization in assigning subjects to different treatments is unimportant in the design of experiments.
10. True or False: The five-digit number 24,516 has 6 integers from 1 to 9 that are its divisors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Carlos Montado was born on Saturday, November 9, 2002. Calculate the day of the week he will be 706 days old and explain your reasoning and the steps you took to find the answer.
12. Dawn is preparing to fix the windows in her house and needs to purchase materials. Calculate the total cost of 3 feet of wood at 7 per foot and 4 pieces of glass at 23 each. Explain your reasoning.

End of Paper